1.1.6 Underlying Philosophy: Introduction to the 1990 Syllabus

This section contains the introduction to the first MEI Structured Mathematics specification and is included as a statement of underlying philosophy. These were the first modular A Levels in any subject and their development was accompanied by serious consideration of how the needs of industry and adult life could best be addressed through a mathematics specification.

‘Our decision to develop this structure, based on 45-hour Components, for the study of Mathematics beyond GCSE stems from our conviction, as practising teachers, that it will better meet the needs of our students. We believe its introduction will result in more people taking the subject at both A and AS, and that the use of a greater variety of assessment techniques will allow content to be taught and learnt more appropriately with due emphasis given to the processes involved.

Mathematics is required by a wide range of students, from those intending to read the subject at university to those needing particular techniques to support other subjects or their chosen careers. Many syllabuses are compromises between these needs, but the necessity to accommodate the most able students results in the content being set at a level which is inaccessible to many, perhaps the majority of, sixth formers. The choice allowed within this scheme means that in planning courses centres will be able to select those components that are relevant to their students' needs, confident that the work will be at an appropriate level of difficulty.

While there are some areas of Mathematics which we feel to be quite adequately assessed by formal examination, there are others which will benefit from the use of alternative assessment methods, making possible, for example, the use of computers in Numerical Analysis and of substantial sets of data in Statistics. Other topics, like Modelling and Problem Solving, have until now been largely untested because by their nature the time they take is longer than can be allowed in an examination. A guiding principle of this scheme is that each Component is assessed in a manner appropriate to its content.

We are concerned that students should learn an approach to Mathematics that will equip them to use it in the adult world and to be able to communicate what they are doing to those around them. We believe that this cannot be achieved solely by careful selection of syllabus content and have framed our Coursework requirements to develop skills and attitudes which we believe to be important. Students will be encouraged to undertake certain Coursework tasks in teams and to give presentations of their work. To further a cross-curricular view of Mathematics we have made provision for suitable Coursework from other subjects to be admissible.

We believe that this scheme will do much to improve both the quantity and the quality of Mathematics being learnt in our schools and colleges.'